

The Law of Large Number

▮ Motivation question:

- We are given a set of identically and independently distributed data / copies, $X_1, \dots, X_n$, of X .
- We do NOT know $X \sim F$.
- We want to estimate $E(X)$.
- Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ be the sample average.
- Why is $\bar{X}_n$ a good choice?

▮ THM Suppose that $X_1, \dots, X_n$
are i.i.d. sample of X , with
 $E(X) < \infty$. Then

$$P\left(\lim_{n \rightarrow \infty} |\bar{X}_n - E(X)| \leq \varepsilon\right) = 1$$

where $\varepsilon > 0$ can be arbitrarily
small.

▮ $\bar{X}_n$ is an asymptotically precise estimator
of $E(X)$.